

# From Gene to Protein: Watching Transcription and Translation

PhET Gene Expression Essentials · PhET

Year 10

~30 minutes

Science · Biology

## NOTICE — AI-generated worksheet

This worksheet was drafted automatically from the site's curriculum-mapping data and has not been checked by a human curriculum expert. Please review every task for accuracy, safety and suitability for your specific class before use.

Simulation: <https://phet.colorado.edu/en/simulations/gene-expression-essentials>

Curriculum links: AC9S10U01 · DNA · genetics

## Learning intentions

- I can describe the roles of RNA polymerase and ribosomes in gene expression.
- I can explain, in order, the main steps of transcription and translation.
- I can explain how gene regulation allows different cell types to use the same DNA differently.

## Getting started

Open the simulation and start on the Transcription screen. Watch one full run of RNA polymerase moving along the DNA before you begin the tasks, then move on to the Translation screen, and finish on the Multiple Cells screen.

## Tasks

1. On the Transcription screen, name the molecule that binds to the DNA to begin transcription and describe what it does as it moves along the gene.

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2. What molecule is produced by the end of transcription, and where does it go next?

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3. Switch to the Translation screen. Describe the roles of the ribosome and tRNA, and explain in your own words how the sequence on the mRNA determines the order of amino acids added to the protein.

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4. Pause the simulation partway through translation. What would happen to the final protein if translation stopped at this point?

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5. Switch to the Multiple Cells screen. Observe protein production in a couple of different cell types over time and record what you see.

| Cell type | Protein level observed |
|-----------|------------------------|
|           |                        |
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6. Based on your Multiple Cells observations, explain why two cells with identical DNA can end up producing very different amounts of a protein.

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7. In your own words, define 'gene expression' using what you've observed in this simulation.

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**Extension (optional)**

Liver cells and skin cells contain exactly the same DNA but look and behave very differently. Using what you learned about gene regulation on the Multiple Cells screen, explain how this is possible.

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